

LESSON GUIDE

Technology Plus

Unit 5: Software

2.6.1 – Virtualization and Cloud Technologies

Lesson Overview:

In this lesson, students will discover virtualization and cloud resources available to assist organizations.

Students will:

- Understand the necessary components of cloud technologies.
- Plan cloud solutions for an organization based on needs and budget.

Guiding Question: In what ways can organizations leverage virtualization and cloud technologies?

Suggested Grade Levels: 8-12

Technology Needed: No

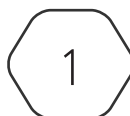
Approximate Time Required: 40-60 minutes

Standards:

CompTIA Tech+ FC0-U71 Objective

- 2.6 – Compare and contrast virtualization and cloud technologies.
 - Virtualization
 - Hypervisor
 - Guest operating system (OS)
 - Cloud concepts
 - Platform as a Service (PaaS)
 - Infrastructure as a Service (IaaS)
 - Software as a Service (SaaS)
 - Deployment models
 - On premises
 - Cloud
 - Hybrid

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CYBER.ORG
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Lesson Background

Background Information:

Students are likely familiar with the cloud and how they interact with it for storage. The purpose of this lesson is to understand how the cloud works and the different requirements, models, and services available. The type of cloud we interact with most often is a public, Software as a Service (SaaS).

Materials Needed:

Materials per Class	• Lesson Guide • Lesson Slides 2.6.1 – Virtualization and Cloud Technologies • Activity Slides 2.6.1 – Building the Cloud on a Budget • Activity 2.6.1 – Company Prospectuses (precut for one per group)
Materials per Group	• Activity 2.6.1 – Cloud Store Handout
Materials per Student	• Student Handout 2.6.1 – Virtualization and Cloud Technologies • Student Notes 2.6.1 – Virtualization and Cloud Technologies • Assessment 2.6.1 – Virtualization and Cloud Technologies (Optional)

Lesson Background

Cyber Terms:

Term	Definition
virtualization	technology that lets you have more than one operating system on the same computer at the same time
cloud computing	the method of accessing remote servers; computing resources shared over a network from remote servers
guest OS	an operating system running on a virtual machine (VM) rather than the host (physical) machine
hypervisor	a piece of software that manages virtual machines
Infrastructure as a Service (IaaS)	network capacity available over the internet
Platform as a Service (PaaS)	virtual hardware (IaaS) and software development tools available over the internet
Software as a Service (SaaS)	programs available over the internet

2.6.1 – Virtualization and Cloud Technologies

Guiding Question: In what ways can organizations leverage virtualization and cloud technologies?

Lesson Launch (3-5 minutes)

- Virtualization and cloud technologies allow for greater flexibility when purchasing computer hardware.
- In the past, a company – or individual – might need a separate computer for each OS they wished to run. Virtualization provided a more cost-effective way to run multiple operating systems on one physical device.
- Cloud computing similarly allows users to ‘rent space’ on servers rather than purchasing enough servers to host all their storage and computing needs.

Home on the Range

- By this point, your students should be comfortable navigating the CYBER.ORG Range, which allows them to run different operating systems using virtual machines (VMs) in a controlled environment.
 - When we navigate to apps.cyber.org, we are accessing and connecting to a Platform as a Service (PaaS).
 - Launching a VM is an example of us interacting with the PaaS.
 - Connecting to the VM shows us connecting to the Infrastructure as a Service (IaaS).
 - Additionally, any time we have used CTRL+ALT+SHIFT to copy and paste from our host OS to our VM demonstrates how the VM acts as a guest OS.

Guests in the Machine (15-20 minutes)

Distribute Student Handout 2.6.1 – Virtualization and Cloud Technologies and encourage students to fill in answers as you go through today’s lesson.

Virtualization lets a single computer run multiple operating systems simultaneously. It provides a more cost-effective way to run multiple operating systems on one physical device.

Requirements

Virtualization requires a host computer, a hypervisor, and a guest OS.

- **Hypervisors**
 - A Type 1 hypervisor is also called ‘bare-metal’ because it is software that works directly on a computer without a main OS.
 - A Type 2 hypervisor works like any other computer program and needs an OS to run.
 - A guest OS runs inside the virtual environment – like the different systems used in labs in the CYBER.ORG Range.

Why Virtualize?

- Uses for virtualization
 - Virtual labs – like many of the labs we’ve done up to this point, we can use a guest OS to create a contained area for deploying and analyzing viruses without infecting the host computer.
 - Legacy software – sometimes we update computer hardware but still need access to older software; VMs let us continue to run older programs even if our host computer no longer supports the software.
 - Development – when new software is created, developers would rather test it in a guest OS where issues can be identified and resolved without impacting the host computer.
 - Training – as we saw in our last lab, VMs let us learn different operating systems without requiring installs on multiple devices; since the CYBER.ORG Range is non-persistent, it also makes it easier for different students to share accounts without interference.

The Cloud

- “as a Service” Models
 - **Infrastructure as a Service (IaaS)** – network capacity available over the internet; can include data center operations, networking components, servers and server hardware, and virtualization software
 - **common uses:** hosting websites or applications, storing data and backups, running high-performance computing tasks
 - **Platform as a Service (PaaS)** – virtual hardware (IaaS) and software development tools available over the internet; can include operating systems and infrastructure software
 - **common uses:** web application development and deployment, manage Application Programming Interfaces (APIs), create mobile backends
 - **Software as a Service (SaaS)** – programs available over the internet; can include hosted applications
 - **common uses:** business productivity tools, customer relationship management, communication and collaboration solutions

Deployment Models

- **On-Premise/Private** – hosting cloud services locally; customizable and more secure
- **Public** – user-friendly because vendors host and manage resources
- **Hybrid** – using both private and public cloud services depending on data being stored, information being accessed, or users needing the products
- **Community** – sharing cloud services amongst a few organizations with similar profiles and common usage needs

Activity: 2.6.1 – Building the Cloud on a Budget (15-30 minutes)

- Divide students into groups of 2–4. Randomly assign each group a Company Prospectus and provide them with the Cloud Store handout.
- Groups will review the needs and budget of their assigned company and select cloud components (servers, hypervisors, and “as a Service” models) to build a customized solution that fits the budget.
- Each group will complete the response section in their Student Handout, where they’ll record:
 - The components they selected.
 - A brief explanation of why those components meet the company’s needs:
- Groups may briefly present their choices to the class, or the instructor may facilitate a discussion where groups compare and explain their decisions.
- If time allows and classroom logistics support it, teachers may optionally allow students to present their proposals in a different format (e.g., spreadsheet, slide deck, or short written pitch).
- Below are examples of solutions groups may have developed.
 - Company ABC – public servers, SaaS
 - Company DEF – 1 private server, public server, PaaS and SaaS
 - Company JKL – public servers, IaaS networking tools, PaaS, and SaaS
 - Company MNP – private servers, SaaS, Type 2 Hypervisors for remote employees
 - Company XYZ – private servers, Type 1 Hypervisors, IaaS, PaaS, and SaaS

Putting It All Together (3-5 minutes)

Discussion Questions

- Which cloud service provides the highest level of abstraction for the client?
Software as a Service
- What is a Guest OS?
An operating system that is running within the main operating system. A virtual OS.
- Imagine you manage a team of software developers for a defense contractor. Describe what type of cloud technologies and what deployment model you would recommend setting up?
This team would want on-premises cloud computing to host IaaS (firewalls), PaaS (development tools), and SaaS technologies.